

SARAH LIBANORE

Postdoctoral Fellow & Science Communicator



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Ben Gurion University
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github.com/slibanore



<https://slibanore.github.io>

RESEARCH INTERESTS

Postdoctoral researcher in astrophysics and cosmology, focused on new observables and cross-correlations. Currently working on line-intensity mapping to study the epoch of reionization and cosmic dawn, with past experience in gravitational waves and large-scale structure tracers. Passionate about teaching, mentoring students, collaborative research, and outreach.

PROFESSIONAL EXPERIENCE

Ben Gurion University
01.2023 - ongoing

Azrieli International Postdoctoral Fellow

Advisor: Prof. Ely D. Kovetz

Research Experience: line intensity mapping, simulation and forecast of 21-cm and star forming lines in the context of current surveys

University of Padova
03.2022 - 12.2022

Pre-Doctoral Research Grant

Advisor: Prof. Michele Liguori

EDUCATION

University of Padova
09.2018 - 05.2022

Ph.D. in Physics

Advisors: Prof. Michele Liguori, Prof. Alvise Raccanelli

Thesis: *The importance of clustering analysis in future GW surveys*

University of Padova
09.2016 - 09.2018

Master degree in Astronomy (110/110 cum laude)

Advisors: Prof. Michele Liguori, Prof. Nicola Bartolo

Thesis: *Gravitational Wave detections: implications for Cosmology*

University of Padova
09.2013 - 09.2016

Bachelor degree in Astronomy (110/110 cum laude)

Advisor: Prof. Sabino Matarrese

TEACHING & MENTORING

- **Tutorial on “21cmFirstCLASS” (Santander, Spain, February 2025)**
- **Assistant Lecturer for “Cosmic Tensions” Ph.D. course (GGI school, Italy, March 2023)**
- **Co-supervisor of of Ph.D. and Master students in their research projects**

Karin Finish (fnl effect on 21-cm at EoR)

Tal Kadosh (island model of EoR)

Keren Krochek (SMBH mass function across z)

Pasquale Tiziano Ursino (fnl effect on GW surveys)

Gali Shmueli (2023, neutrino masses and LIM)

Alberto Magaraggia (2022, SGWB from PBH)

Matteo Peron (2021, bias of GW mergers)

GRANTS & AWARDS

- **BGU/Philadelphia academic bridge grant (2025)**
Funds 2 months visit to UPenn (Philadelphia) in collaboration with Prof. Adam Lidz
- **Swiss National Science Foundation Scholarship (2024)**
Funds attendance to Cosmology in the Alps Conference in Les Diablerets (Switzerland)
- **Balzan award held by Center for Cosmological Studies (2023)**
Funds 1 month visit to JHU (Baltimore) in collaboration with Prof. Marc Kamionkowski
- **Azrieli International Fellowship program (2022)**
- **Fondazione Aldo Gini scholarship (2022)**
Funds 6 months visit to Ben Gurion University (Israel) in collaboration with Prof. Ely D. Kovetz
- **PhD Sandwich Scholarship, held by Council for Higher Education in Israel (2022)**
Funds 6 months visit to Ben Gurion University (Israel) in collaboration with Prof. Ely D. Kovetz
- **Mille e una Lode, held by University of Padova (2016)**

TALKS & POSTERS (MAIN AND MOST RECENT CONFERENCES)

- Annual meeting of the Israely Physical Society (Haifa, Israel, July 2025)
- LIM 2025 (Annecy, France, June 2025)
- SUBLIME meeting (Tokyo, Japan, March 2025)
- 21 cm Cosmology Workshop (Hangzhou, China, July 2024)
- LIM 2024 (Urbana Champaign, USA, May 2024)
- Cosmology in the Alps (Les Diablerets, Switzerland, March 2024)
- HI as a Cosmological probe (Nazareth, Israel, May 2023)

SEMINARS

- Santander, Spain, February 2025
- Bar Ilan University, Israel, February 2024
- Scuola Normale Superiore di Pisa, Italy, November 2023

COLLABORATIONS & MEMBERSHIPS

- Square Kilometer Array Observatory (SKAO)
- International Astronomical Union (IAU)
- European Consortium for Astroparticle Theory (EUCAPT)
- ULTRASAT

REFERENCES

- **Ely D. Kovetz**
Ben Gurion University
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- **Julian B. Muñoz**
University of Texas, Austin
julianbmunoz@utexas.edu
- **Michele Liguori**
Università degli studi di Padova
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- **Alvise Raccanelli**
Università degli studi di Padova
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LIST OF PUBLICATIONS

FIND MY WORK
ON INSPIRE HEP



- **S. Libanore, E.D. Kovetz, J.B. Muñoz, A.Lidz (2025)**
arXiv:2509.XXXX – *A New Boundary Condition on Reionization*
- **S. Libanore, J.B. Muñoz, E.D. Kovetz (2025)**
arXiv 2507.15922 – *oLIMpus: An Effective Model for Line Intensity Mapping Auto- and Cross- Power Spectra in Cosmic Dawn and Reionization*
- **S. Libanore, S. Ghosh, E.D. Kovetz, K.B. Body, A. Raccanelli (2025)**
arXiv:2504.15348 – *Joint 21-cm and CMB Forecasts for Constraining Self-Interacting Massive Neutrinos*
- G. Shmueli, **S. Libanore, E.D. Kovetz (2024)**
Phys. Rev. D 111 063512 (2025), **arXiv:2412.04071** – *Towards a multi-tracer neutrino mass measurement with line-intensity mapping*
- J. Flitter, **S. Libanore, E.D. Kovetz (2024)**
Phys.Rev.D 112 023537 (2025), **arXiv:2412.04071** – *More careful treatment of matter density fluctuations in 21-cm simulations*
- E. Vanzan, **S. Libanore, L. Valbalsa Dall'Armi, N. Bellomo, A. Raccanelli (2024)**
JCAP 10 014 (2024), **arXiv:2405.13871** – *Gravitational wave background from primordial black holes in globular clusters*
- **S. Libanore, E.D. Kovetz (2024)**
Phys. Rev. D 111 063521 (2024), **arXiv:2404.01500** – *Upcoming searches for decaying dark matter with ULTRASAT ultraviolet maps*
- **S. Libanore, E.D. Kovetz (2024)**
A&A 687, A133 (2024), **arXiv:2401.12285** – *Constraining $z<2$ ultraviolet emission with the upcoming ULTRASAT satellite*
- **S. Libanore, J. Flitter, E.D. Kovetz, Z. Li, A. Dekel (2023)**
MNRAS 532 1 (2024), **arXiv:2310.03021** – *Effects of feedback-free starburst galaxies on the 21-cm signal and reionization history*
- **S. Libanore, M. Liguori, A. Raccanelli (2023)**
JCAP 08 055 (2023), **arXiv:2306.03087** – *Signatures of primordial black holes in gravitational wave clustering*
- M. Peron, A. Ravenni, **S. Libanore, M. Liguori, M.C. Artale (2023)**
MNRAS 530 1 (2024), **arXiv:2305.18003** – *Clustering of binary black hole mergers: a detailed analysis of the EAGLE+MOBSE simulation*
- T. Adi, **S. Libanore, H.A. Cruz, E.D. Kovetz (2023)**
JCAP 09 035 (2023), **arXiv:2305.06440** – *Constraining Primordial Magnetic Fields with Line-Intensity Mapping*
- **S. Libanore, C. Unal, D. Sarkar, E.D. Kovetz (2022)**
Phys. Rev. D 106 123512 (2022), **arXiv:2208.01658** – *Unveiling cosmological information on small scales with line intensity mapping*
- **S. Libanore, M.C. Artale, M. Liguori, M. Mapelli et al. (2021)**
JCAP 02 003 (2022), **arXiv:2109.10857** – *Clustering of Gravitational Wave and Supernovae events: a multitracers analysis in Luminosity Distance Space*
- **S. Libanore, M.C. Artale, M. Liguori, M. Mapelli et al. (2020)**
JCAP 02 035 (2021), **arXiv:2007.06905** – *Gravitational Wave mergers as tracers of Large Scale Structures*

OUTREACH & DISSEMINATION

PLaNCK! magazine
2016 - ongoing

Editorial Board of science magazine for children

Article writing and editing
Collaboration in public relations and social media management

APS Accatagliato
2016 - ongoing

Outreach activities tailored to diverse audiences

Planning, design, and implementation of workshops, festivals, and events
Science trainer & educator in schools, workshops, professional internships
Back-office support activities

Penne di Scienza
2024 - ongoing

Correspondence with school classes

Engaged with teenagers via letters and online meetings to foster science communication and provide role modeling

COMPETENCIES

Personal & Soft Skills

Leadership & Management
Teamwork & collaboration
Communication & active listening
Organization & planning (for self and others)
Big-picture thinking & summarizing
Adaptability & stress management
Cultural awareness & open-mindedness
Language learning & multilingualism
Travel planning & curiosity

Professional & Technical Skills

Scientific software
Programming: Python, C++
Computational problem solving
Data visualization & scientific plotting
Code documentation & reproducibility
Statistical methods
Scientific writing & research communication
Public speaking & presentation design
Graphic design (basic)